Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/12/2019
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	Circulating miRNAs Provides Systemic Host Response to Microgravity: Utilizing GeneLab Datasets to Identify Molecular Targets for Spaceflight Studies		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2021 per F. Hernandez/ARC (Ed., 10/20/20)		
Key Personnel Changes/Previous PI:	Ed. note 11/6/2019: CoInvestigators have changed per approval of NASA Ames--Dr. Peter Grabham replaces original CoI Dr. John McDonald and Dr. Elge Cekanaviciute is added as CoI to assist Dr. Charles Vanderburg who was a previous Co-I and now is a consultant on the grant.		
COI Name (Institution):	Blaber, Elizabeth Ph.D. (Rensselaer Polytechnic Institute grant 80NSSC20K0307 (formerly NASA ARC employee)) Grabham, Peter Ph.D. (Columbia University grant 80NSSC19K1658) Cekanaviciute, Egle Ph.D. (Universities Space Research Association grant 80NSSC19K0883/NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Biological risks associated with microgravity are a major concern for long-term space travel. Although determination of risk has been a focus for NASA biological research, data examining systemic (i.e., multi- or pan-tissue) responses to spaceflight are sparse. For this proposal, we utilized the NASA GeneLab database to analyze the transcriptomic data from 7 different rodent tissue datasets. Using our systems biology expertise, we were able to reveal for the first time the existence of potential "master regulators" coordinating systemic responses to microgravity. We identified p53, Transforming Growth Factor Beta 1 (TGFB1), and immune related pathways as the highly prevalent pan-tissue signaling pathways that are affected by microgravity responses. Key genes/drivers isolated by our unbiased systems biology analyses displayed a major overlap between tissue types and established that TGFB1 was the most inter-connected gene across all datasets. Finally, a hypothesis was generated indicating that an overall systematic microgravity response driven by TGFB1 is caused by a circulating microRNA (miRNA) signature. This miRNA signature, consisting of 13 significantly expressed miRNAs, was predicted from differential gene expression that causes miRNA-driven microgravity-related functional states. A subsequent impact on health using a theoretical "health risk score" was calculated. The genes and miRNAs identified from our analyses will be used for the ground based studies which will be proposed for this solicitation and will be investigated to develop potential countermeasures against microgravity response. This proposal exemplifies the utility of the GeneLab data repository to aid in the process of performing novel hypothesis-based spaceflight research aimed at elucidating the global impact of environmental stressors at multiple biological scales.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This miRNA signature can be a minimally invasive biomarker to monitor health risk during deep space missions. The antagomir countermeasures being developed can act as a preventive measure against space radiation and microgravity effects. Countermeasures can be applied throughout mission to potentially mitigate any space environment effects. The same techniques can be used in the clinic to monitor various diseases such as muscle degeneration risk in patients. miRNA signature can be used during yearly checkup as another diagnostic blood based test to monitor risk for these diseases. If miRNA signature starts to appear in patients, then can apply antagomir treatment to suppress active miRNAs that are associated with disease progression long before the manifestation of the disease will occur.
Task Progress:	There are many known health risks currently associated with space travel with the majority of the risk stemming from two components: microgravity and radiation. From our earlier work, we predicted that there is a systemic component of the host that causes general increased health risks due to spaceflight driven by a circulating microRNA (miRNA) signature consisting of 13 miRNAs that directly regulates both p53 and TGFß1 and we have also identified a specific 9 miRNA signature that detects accurately the presence of lymphoma. MiRNAs are small non-coding RNAs that are known to circulate stably throughout the body and have been found in the majority of the bodily fluids. We hypothesize that using our technique there should be an identifiable unique miRNA signature for each specific disease. We have started to dissect the actual impact of this miRNA signature on both the radiation and microgravity components and prove that this miRNA signature actually exists in the circulation of a host impacted by spaceflight. To achieve this, we obtained multiple tissues (serum, liver, and spleen) from C57BL/6 male mice that were hindlimb unloaded (HU) to simulated microgravity, irradiated with 2Gy gamma (IR), HU plus IR, and control mice. In addition, we have also obtained serum and tissues from mice with HU and irradiated with 1Gy iron, 1Gy proton, and a mix beam of 0.5Gy Proton and 0.5Gy iron. We have run ddPCR on the serum from the archived serum that was gathered from various experiments in the past from Drs. Ruth Globus and Ann-Sofie Schreurs. We completed the miRNA ddPCR profiling on all the serum from the above experiments. The data shows amazingly that my initial in silico predictions for the miRNA signature that were the basis of my hypothesis, do indeed exist in serum of mice exposed to simulated spaceflight conditions. The archived tissues that were used also had gamma irradiated mice both with and without simulated microgravity. The circulating spaceflight miRNA signature was shown to not be associated with gamma. This part of the data shows that this miRNA signature is indeed truly associated with spaceflight conditions and does not appear with other conditions. We further went to characterize the functional impact of the miRNAs. This was done through in silico predictions to determine the key miRNA targets and pathways that the significant spaceflight miRNA signature will impact. We show that the key genes that are targeted by the significantly regulated miRNAs are related to known spaceflight associated genes. For example, the targeted genes related to cell cycle, such as CDKN2A and CDK2, which are known to be dysregulated in many tissues due to the stressors of spaceflight. The diseases that predicted to be regulated by the spaceflight associated miRNA signature are all known health risks that are associated with spaceflight. The miRNAs seem to be involved in many types of cancer and increased cancer risk is known to occur due to exposure to the space environment. In addition, Central Nervous System (CNS) diseases, cardiovascular diseases, and muscle degeneration are predicted to be regulated by the increase of these miRNAs. Lastly, many familiar pathways, such as TGFß, p53, and cell cycle, are being significantly targeted by these miRNAs. This data shows that this confirmed spaceflight associated miRNA signature is targeting many known diseases, pathways, and genes associated the increased health risks due to spaceflight. The potential of utilizing this miRNA signature as a methodology to monitor health risks in long term space missions would be beneficial for astronaut's health. We have also started to add additional data from archived serum from mice that were flown to the International Space Station (ISS). This tissue has been awarded to Dr. Beheshti from two CASIS (Center for the Advancement of Science in Space) tissue grants (RRRM-1 and RRRM-2). This data will add the component of actual spaceflight samples to confirm what is being observed with the simulated microgravity and space radiation experiments. To confirm if this miRNA signature actually exists in humans, we have started to add additional data in collaboration with Dr. Sylvain Costes utilizing human peripheral blood mononuclear cells (PBMCs) that are exposed to space radiation (i.e., Iron particles irradiated at Brookhaven National Laboratories). N=12 human PBMCs were irradiated with the following conditions: 1) 0Gy, 0.3Gy of 56Fe, and 3) 0.82Gy of 56Fe. We quantified the miRNAs from these PBMCs 4 hours after irradiation. With this data we see an interesting response with the miRNAs, with the miRNAs increase in amount of expression in the PBMCs at 0.3Gy of 56Fe, but the majority seem to decrease in expression back to the 0Gy level at 0.82Gy of 56Fe. The decrease at the higher dose for the PBMCs is attributed to the higher cell death at the higher doses, which is observed with the PBMCs in culture. We are currently doing further analysis to quantify the cell death at the higher doses and normalize the data to account for this observation. Qualitatively we account this reduction of miRNAs at higher doses to the increased cell death. The overall trend before cell death at 0.3Gy for the PBMCs is still in agreement with the overall results from the serum in mice. This data indicates that even a healthy population of

	human PBMCs when exposed to space radiation will activate the spaceflight associated miRNAs in isolate cells. One can imagine continuous stress of spaceflight for long-term missions will create a continuous expression of miRNAs in the circulation and contribute to the spaceflight associated health risks. We will also be adding additional miRNA data from the NASA Twin Study. Dr. Beheshti has been collaborating with Dr. Chris Mason on analyzing the miRNA data associated with NASA Twin Study. From this analysis it was revealed that this miRNA signature does indeed exist in the astronaut who was on the ISS for a year, while it does not exist in the controls. This data can't be shown until we obtain the proper approvals from Johnson Space Center (JSC). Once we have a complete paper draft for this first part of the project, we will send all figures and paper draft to JSC for proper approvals. Finally, functional assays have started on in vitro cell culture by manipulating these miRNAs. This involved both promoting expression of the miRNAs and inhibiting miRNAs in various cell lines to either mimic spaceflight risk or mitigate the microgravity effects. This is currently being conducted by Co-Investigator Dr. Peter Grabham at Columbia University. We are planning on submitting a manuscript with all these results to Science by the end of February which is close to the first-year completion of the grant. I believe we have successfully been progressing with the aims of our proposal and have gone far and beyond with what has been proposed for the first year of the grant. Here we will provide a bullet list of accomplishments and progress to outline clearly the progress being made with this grant: • Accomplishments: o Characterized predicted spaceflight miRNA signature in archived serum from mice under both simulated microgravity and space radiation experiments. o Added additional human data to confirm that the miRNA signature and results translates to humans. o This data indicates potential use of the miRNAs as a decent biomarker to monitor health risks for space missions. o Manuscript in preparation to be submitted to Science by the end of February 2020. • PI's achievements from this work: o A provisional patent is filed and discussions have started to potentially filing a full patent based on these results for a methodology to use these miRNAs as a biomarker. o Received the NASA Exceptional Scientific Achievement Medal.
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Awards	Beheshti A. (Afshin Beheshti) "NASA Exceptional Scientific Achievement Medal, August 2019." Aug-2019
Patents	ARC-18235-1. Provisional Patent filed on March 8, 2019. Mar-2019 Beheshti A, McDonald JT, Vanderburg C, Costes SV. "Methods for identifying and treating negative health impacts of exposure to spaceflight environments."